

Corrigé — Exercice 3

$$I : u = \ln x, \, dv = x \, dx : I = \left[\frac{x^2}{2} \ln x \right]_1^e - \frac{1}{2} \int_1^e x \, dx = \frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right) = \frac{e^2 + 1}{4}.$$

$$J : u = x - 1, \, dv = \cos x \, dx : J = [(x - 1) \sin x]_0^\pi - \int_0^\pi \sin x \, dx = 0 - [-\cos x]_0^\pi = -2.$$

$$L : u = x + 2, \, dv = e^{-x} \, dx : L = [-(x + 2)e^{-x}]_0^1 + \int_0^1 e^{-x} \, dx = \left(2 - \frac{3}{e} \right) + \left(1 - \frac{1}{e} \right) = 3 - \frac{4}{e}.$$

$$K : u = x - 2, \, dv = e^{2x} \, dx : K = \left[\frac{x-2}{2} e^{2x} \right]_1^2 - \frac{1}{2} \int_1^2 e^{2x} \, dx = \frac{e^2}{2} - \frac{1}{4} (e^4 - e^2) = \frac{3e^2 - e^4}{4}.$$

$$2) \text{ Première IPP } (u = x^2, v' = e^{-x}) : M = [-x^2 e^{-x}]_0^1 + 2 \int_0^1 x e^{-x} \, dx = -e^{-1} + 2 \int_0^1 x e^{-x} \, dx.$$

$$\text{Deuxième IPP : } \int_0^1 x e^{-x} \, dx = [-x e^{-x}]_0^1 + \int_0^1 e^{-x} \, dx = -e^{-1} + (1 - e^{-1}) = 1 - 2e^{-1}. \text{ D'où}$$

$$M = -e^{-1} + 2(1 - 2e^{-1}) = \boxed{2 - \frac{5}{e}}.$$